

# POLYP DETECTION MEDICAL REPORT

Analysis Date: June 03, 2026

## VIDEO INFORMATION

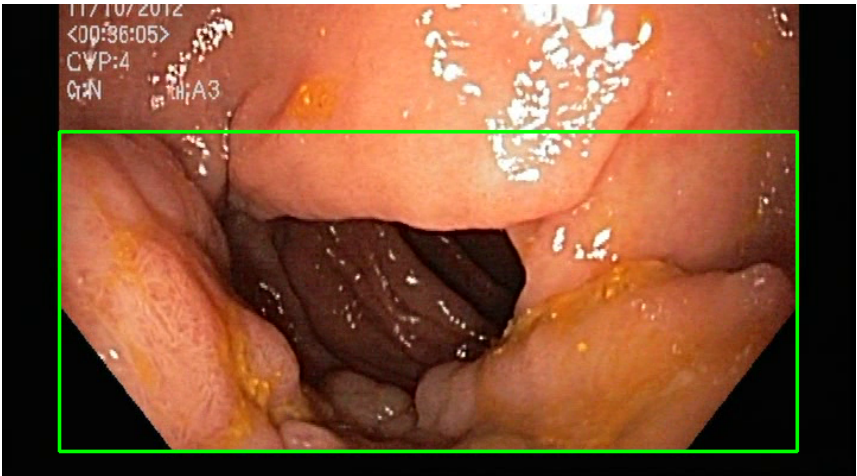
|               |                                      |
|---------------|--------------------------------------|
| Video ID:     | 12ebca74-2378-4498-b2c1-44e0a3ae9f9f |
| Total Frames: | 926                                  |
| Frame Rate:   | 25.0 fps (assumed)                   |
| Duration:     | 37.0 seconds                         |

## DETECTION SUMMARY

| Detection Method              | Count        | Status            |
|-------------------------------|--------------|-------------------|
| YOLO Detections               | 444          | ✓                 |
| RT-DETR Detections            | 596          | ✓                 |
| MedSAM2 Detections            | 611          | ✓                 |
| Consensus (3-Model Agreement) | 1            | ✓                 |
|                               |              |                   |
| <b>Risk Classification</b>    | <b>Count</b> | <b>Percentage</b> |
| HIGH RISK                     | 1            | 100.0%            |
| MEDIUM RISK                   | 0            | 0.0%              |
| LOW RISK                      | 0            | 0.0%              |
| TOTAL ANALYZED                | 1            | 100.0%            |

# DETAILED POLYP ANALYSIS

**POLYP #1 — MALIGNANT\_POLYP [HIGH\_RISK] (Tracked across 310 frames)**



| Feature                 | Value                                    | Interpretation                                                          |
|-------------------------|------------------------------------------|-------------------------------------------------------------------------|
| Frame Index             | 22                                       | Frame 22 of 926                                                         |
| Detection Method        | Detector Consensus (MedSAM2 Unavailable) | MedSAM2 Unavailable (MedSAM2 unavailable for this video)                |
|                         |                                          |                                                                         |
| POLYP TYPE              | MALIGNANT_POLYP                          | Type confidence: 92.0%                                                  |
|                         |                                          |                                                                         |
| Redness Score           | 0.227                                    | Color-based vascularization                                             |
| Relative Radius         | 42.1%                                    | Polyp size as % of frame diagonal                                       |
| Texture Score           | 0.165                                    | Surface morphology                                                      |
| Vessel Visibility       | 0.505                                    | Vascularization indicator                                               |
| Risk Tier               | HIGH RISK                                | Validated by ESGE 2017 / NICE classification                            |
|                         |                                          |                                                                         |
| Clinical Classification | LOW_RISK                                 | Final symbolic reasoning bucket                                         |
| Expert Prediction       | LOW_RISK                                 | Decision Tree output                                                    |
| Clinical Class          | MALIGNANT_POLYP                          | High suspicion for malignancy — biopsy and urgent resection recommended |
| Detection Confidence    | 87.7%                                    | YOLO / RT-DETR / track confidence                                       |

**Type Assessment: Morphology consistent with high-grade neoplasia — biopsy and urgent resection strongly recommended.**

*Low-confidence HIGH RISK detection - Further evaluation recommended*

# CLINICAL IMPRESSION

**Summary:**

A total of 1 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

**Risk Distribution:**

- HIGH RISK polyps: 1 (100.0%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 0 (0.0%)

**Recommendation:**

The presence of HIGH RISK polyps warrants close clinical attention. Consider endoscopic resection or follow-up surveillance depending on clinical context.

**Technical Details:**

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

**Disclaimer:** This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.